

High-Level Design (HLD) Document

Matrimony Platform MVP Architecture & System Specification

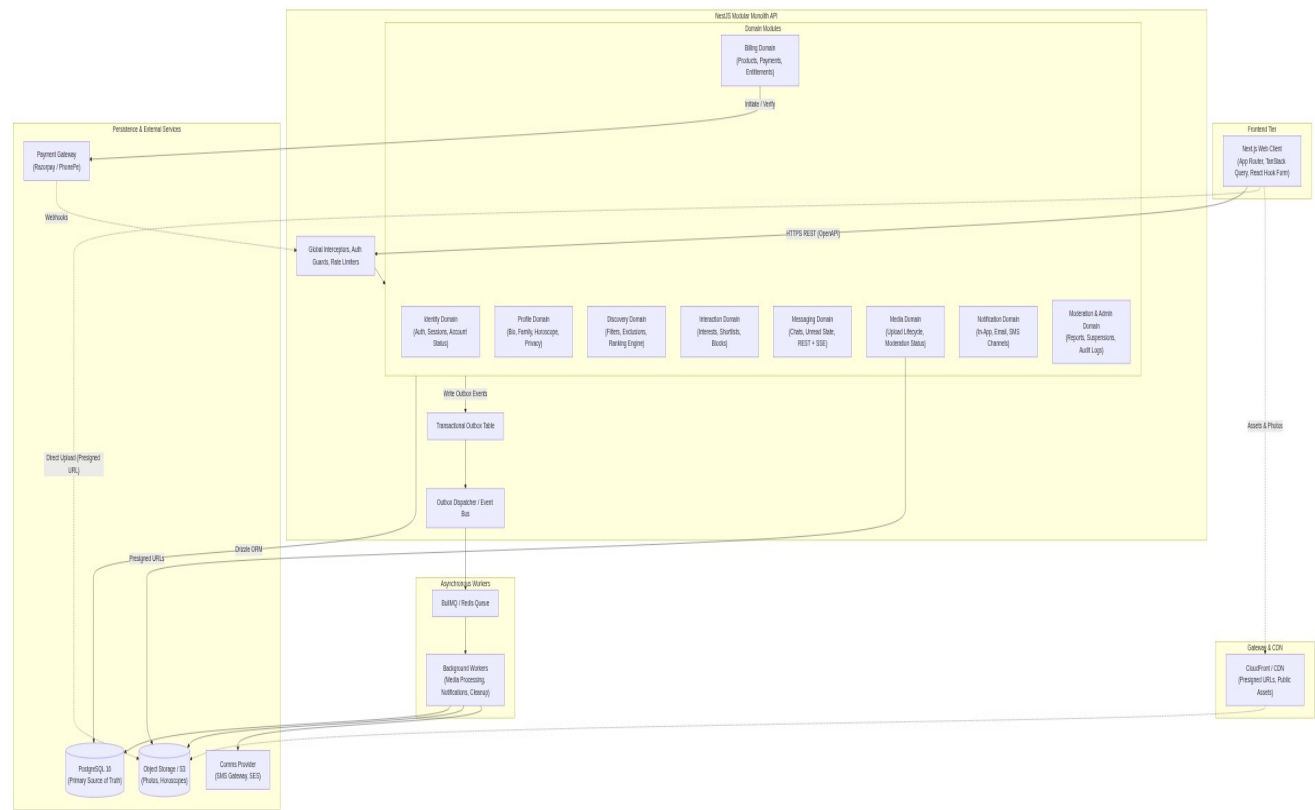
1. Executive Summary & Architectural Principles

This document defines the high-level system architecture for the Matrimony Platform MVP. It establishes explicit domain boundaries, deterministic business scoring algorithms, resilient event processing, and robust domain-driven security models while preventing premature microservices, unconstrained abstractions, or raw file pass-throughs.

- **Domain Capability Boundaries:** The system is decomposed into explicit business domains (Identity, Profile, Discovery, Interaction, Billing, Media, Messaging, Notifications, Moderation, Admin) rather than repetitive technical wrapper layers.
- **Single Contract Authority:** OpenAPI and shared Zod schemas (in packages/contracts) serve as the unified source of truth for both frontend and backend validation.
- **Direct Media Lifecycle:** Client directly uploads binary media to S3 via presigned URLs with asynchronous thumbnail generation, EXIF stripping, and moderation queues.
- **Deterministic Discovery & Ranking:** PostgreSQL-powered multi-factor compatibility ranking (without premature machine learning or Elasticsearch complexity).
- **Resilient Distributed Outbox:** Transactional Outbox pattern guarantees event consistency for asynchronous jobs, notifications, and webhooks.
- **Domain-Driven Privacy & Authorization:** Fine-grained visibility matrix governs contact phone numbers, photo visibility, and mutual blocklists.

2. High-Level System Architecture

The system operates as a modular monolith on the backend with a modern server-rendered React frontend.



3. Monorepo Structure & Domain Boundaries

```
matrimony-app/  
  ■■■ apps/  
    ■ ■■■ web/ (Next.js 14/15, TanStack Query, React Hook Form, Tailwind CSS)  
    ■ ■■■ api/ (NestJS Modular Monolith API with Fastify engine)  
    ■■■ packages/  
      ■ ■■■ contracts/ (Single source of truth: Zod schemas & DTO contracts)  
      ■ ■■■ database/ (Drizzle ORM schema, migrations, connection client)  
      ■ ■■■ config/ (System constants, pricing tiers, domain enums)  
      ■■■ infrastructure/ (Docker, Docker Compose, Deployment specs)
```

4. Core Subsystems & Key Workflows

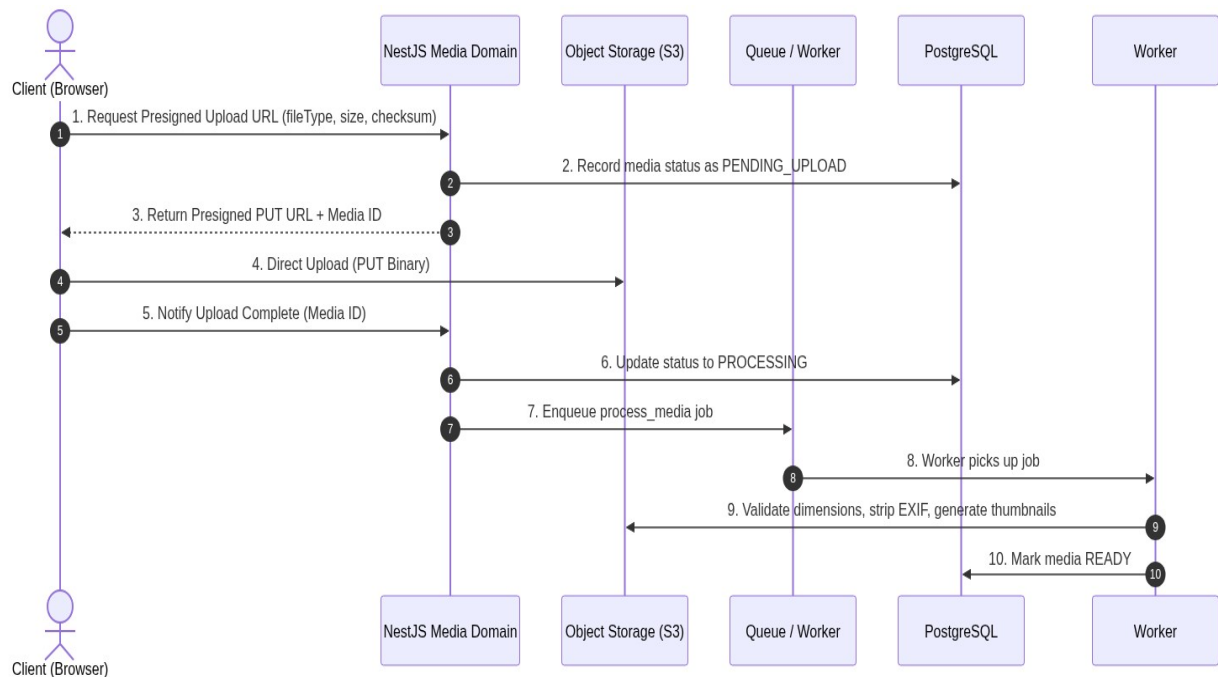
4.1. Discovery & Deterministic Ranking Engine

Discovery combines **hard constraint SQL filtering** with a **deterministic scoring algorithm** to produce relevant matches without ML overhead:

Criteria	Weight / Points	Description
Exact Location / City	+25 pts	Matches candidate partner preference city/district
Education & Occupation	+20 pts	Matches desired qualification bracket
Horoscope / Guna Match	+20 pts	Deterministic astrological alignment matrix score
Profile Completeness	+15 pts	Verified ID, multiple photos, detailed family bio
Recent Activity	+20 pts	Active within last 3 days / 7 days

4.2. Direct-to-Storage Media Lifecycle

Photos are never streamed through the application server. Presigned direct uploads with background worker processing ensure high throughput and security.



4.3. Billing, Payments & Entitlements State Machine

Separates catalog **Products**, financial **Payments**, and granted **Entitlements**. Incoming payment webhooks are processed with idempotent database locks (SELECT ... FOR UPDATE) to guarantee zero double-crediting.

4.4. Reliable Eventing (Transactional Outbox)

All domain side-effects (notifications, SMS alerts, photo thumbnailing, search index updates) are written to an outbox_events table inside the identical database transaction as the business entity, then dispatched asynchronously to BullMQ workers.

5. Abuse Prevention, Rate Limiting & Moderation

Attack / Risk Vector	Protection Mechanism
OTP / SMS Spam	Strict IP + Phone rate limiting. Maximum 3 OTPs per phone number per hour. Cloudflare Turnstile bot verification.
Profile Scraping	Token bucket rate limits on /search and /profiles/:id. Pagination capped to 100 pages.
Harassment / Bulk Messaging	Tiered daily caps on Interests & Messages (e.g. 15/day for free tier). Instant bidirectional block capability.
Fake / Inappropriate Photos	Automated EXIF stripping, dimensions check, and mandatory admin photo moderation queue prior to public display.
Privileged Abuse	Role-Based Access Control (RBAC) with append-only immutable admin_audit_logs.

6. Core Database Schema & Entity Relationships

